ADITYA PAWAR

Computer Science & Engineering Student

Kolhapur, Maharashtra, India | +91 8830537621 | chandrashekharpawar105@gmail.com | GitHub

PROFILE SUMMARY

Final-year Computer Science & Engineering student graduating in 2027 with hands-on experience in full-stack development, Android development, IoT, and on-device machine learning. Proficient in C++, Python, Java, SQL, and modern development frameworks, with a foundation in Data Structures, DBMS, Operating Systems, and Computer Networks. Experienced in building real-time applications, AI-powered mobile solutions, and hardware-integrated systems. Familiar with Git, REST APIs, and structured software development, with strong problem-solving, analytical thinking, adaptability, and willingness to learn new technologies.

EDUCATION

B.Tech - Computer Science & Engineering — **Expected 2027**
D.Y. Patil College of Engineering & Technology, Kolhapur — **77.09%**

Diploma - Computer Engineering — **2021–2024**
D.Y. Patil Polytechnic, Kolhapur — **88.63%**

SSC / Class 10
Shri. V. J. Deshmukh High School — **93.20%**

TECHNICAL SKILLS

Languages: C++, Python, Java, SQL, C **Core:** Data Structures, DBMS, Operating Systems, Computer Networks, Object-Oriented Programming
Development: Android, XML, React, Node.js, Express.js, MongoDB, REST APIs, WebSockets
Machine Learning: TensorFlow, TensorFlow Lite, Keras, On-Device AI **Databases:** Firebase, Firestore, Room Database
Tools: Git, GitHub, Android Studio, VS Code, Arduino, Command Line

PROFESSIONAL SKILLS

Team Collaboration, Communication, Problem Solving, Analytical Thinking, Adaptability, Task Prioritization, Quick Learner, Cross-functional Collaboration

EXPERIENCE

IoT & Embedded Systems Intern — June 2023 – August 2023

- Developed an IoT-based proximity detection system using **Arduino and Raspberry Pi**, achieving up to **95% distance detection accuracy** during real-time testing.
- Programmed real-time distance sensing and alert mechanisms using **C**; integrated sensors with embedded hardware and deployed a functional **public-safety prototype**.

PROJECTS

Drishti AI – Assistive Vision & Hazard Detection App

Android Java , CameraX, TensorFlow Lite, YOLO26, ML Kit Face Detection, MobileFaceNet, Gemini API, Android Text-to-Speech

- Developed an accessible Android app for blind and low-vision users, providing live object recognition and spoken alerts describing detected objects and their location in the camera view.
- Implemented on-device TensorFlow Lite detection with separate object and pothole modes, allowing the app to run one detector at a time.
- Added known-person enrollment and recognition using ML Kit face detection and MobileFaceNet, with spoken name alerts.
- Integrated double-tap scene descriptions the app sends a single camera snapshot to Gemini and displays and speaks the response while camera detection continues.
- Added TalkBack-friendly controls, configurable haptic alerts, and automatic torch controls for low-light use.

Talent IQ - Real-Time Interview Preparation & Coding Platform

React, Node.js, Express.js, MongoDB, WebSocket, REST APIs

- Developed a full-stack platform featuring **mock interviews, coding challenges, an in-browser code editor, and real-time collaboration**.
- Built a multi-language coding environment with live execution for **50+ coding problems** without external setup.
- Implemented authentication, session management, and interview tracking to support **30+ concurrent sessions**.
- Engineered **Node.js**/**WebSocket** communication, improving data synchronization speed by approximately **40%**; integrated automated email and **AI-assisted interview preparation** features.

CropScan - AI-Powered Crop Disease Detection & Advisory System

Android (Java), TensorFlow Lite, Keras, Firebase, CameraX, Room Database, REST APIs

- Ran offline TensorFlow Lite disease detection with Test-Time Augmentation for Banana, Brinjal, Cotton, and Maize.
- Offline Room storage; Firebase Firestore sync; GPS Open-Meteo 7-day weather, severity and treatment advice.